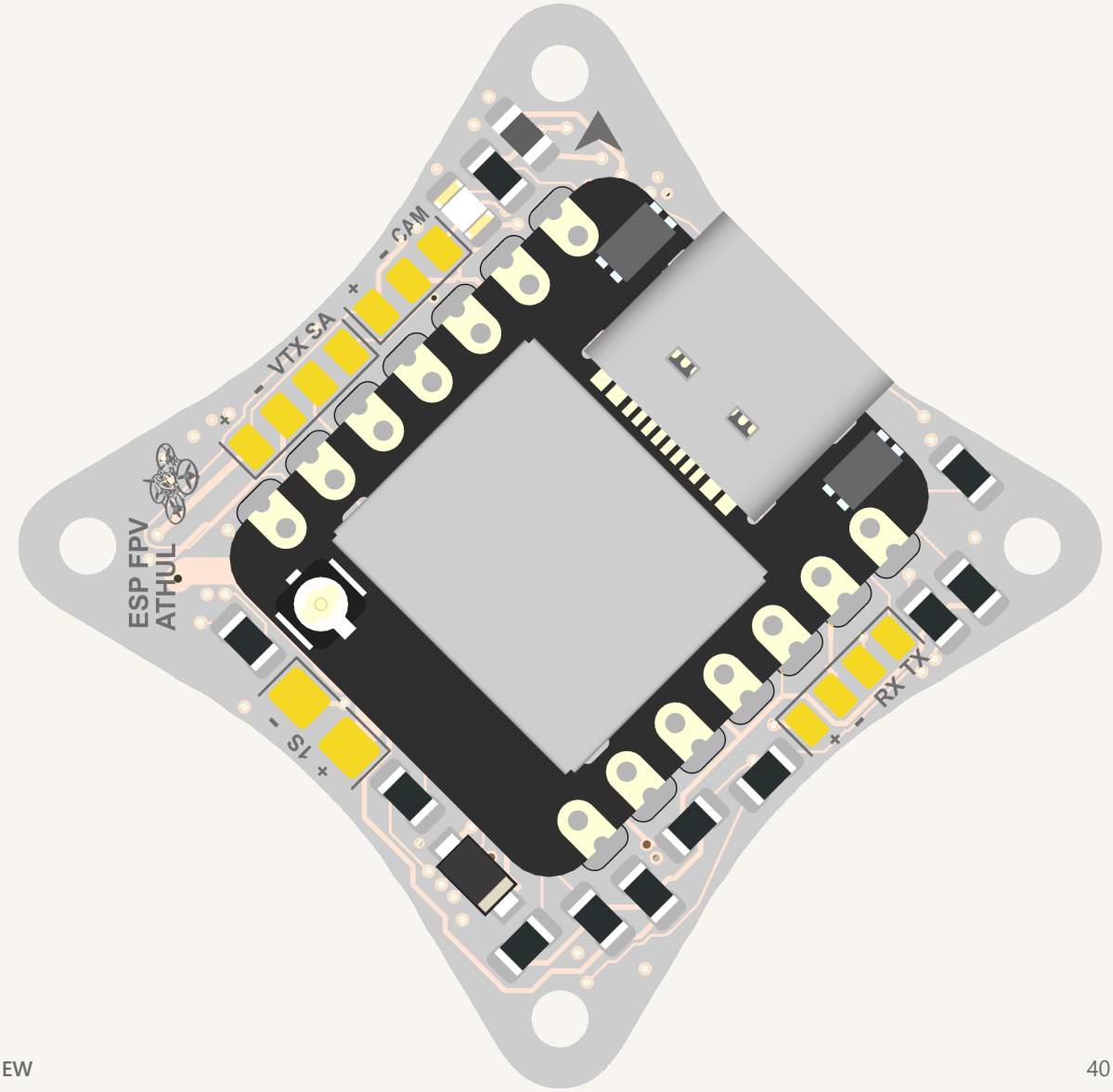
ESP FPV

MATRIX40 / 1S BRUSHED FLIGHT CONTROLLER



FRONT VIEW

40 x 40 mm

USER MANUAL

Connections, configuration and first use

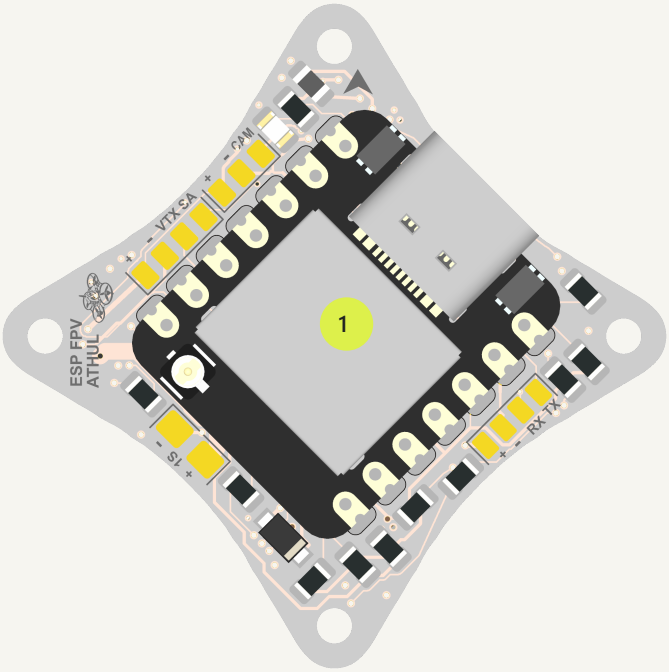
HARDWARE REVISION L4

01

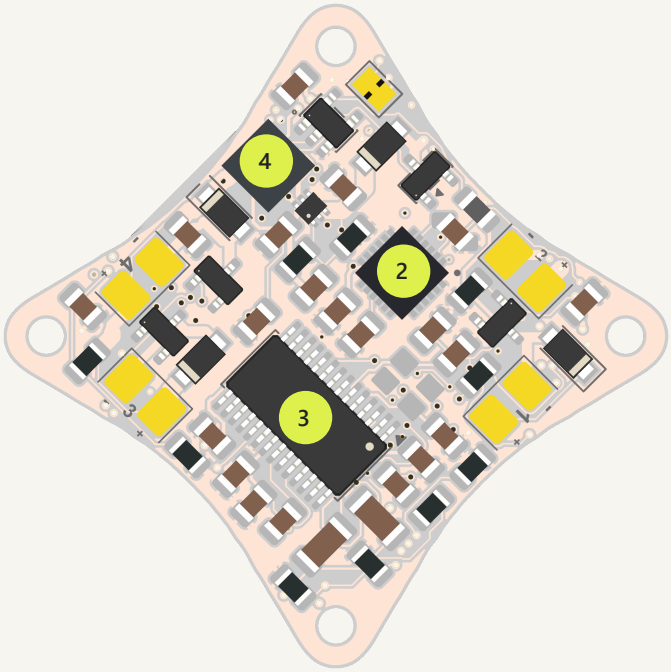
HARDWARE OVERVIEW

THE CONTROLLER

A 1S brushed FPV controller built around the XIAO ESP32-S3, with onboard motion sensing and analog video overlay.



A / FRONT



B / BACK

01 / CONTROL

XIAO ESP32-S3 module. USB connects directly to the module for programming and diagnostics.

02 / MOTION

MPU-6050 accelerometer and gyroscope. Check axis direction in your firmware before flight.

03 / VIDEO

AT7456E analog OSD. Camera video enters J2; the overlay output leaves through J3.

04 / POWER

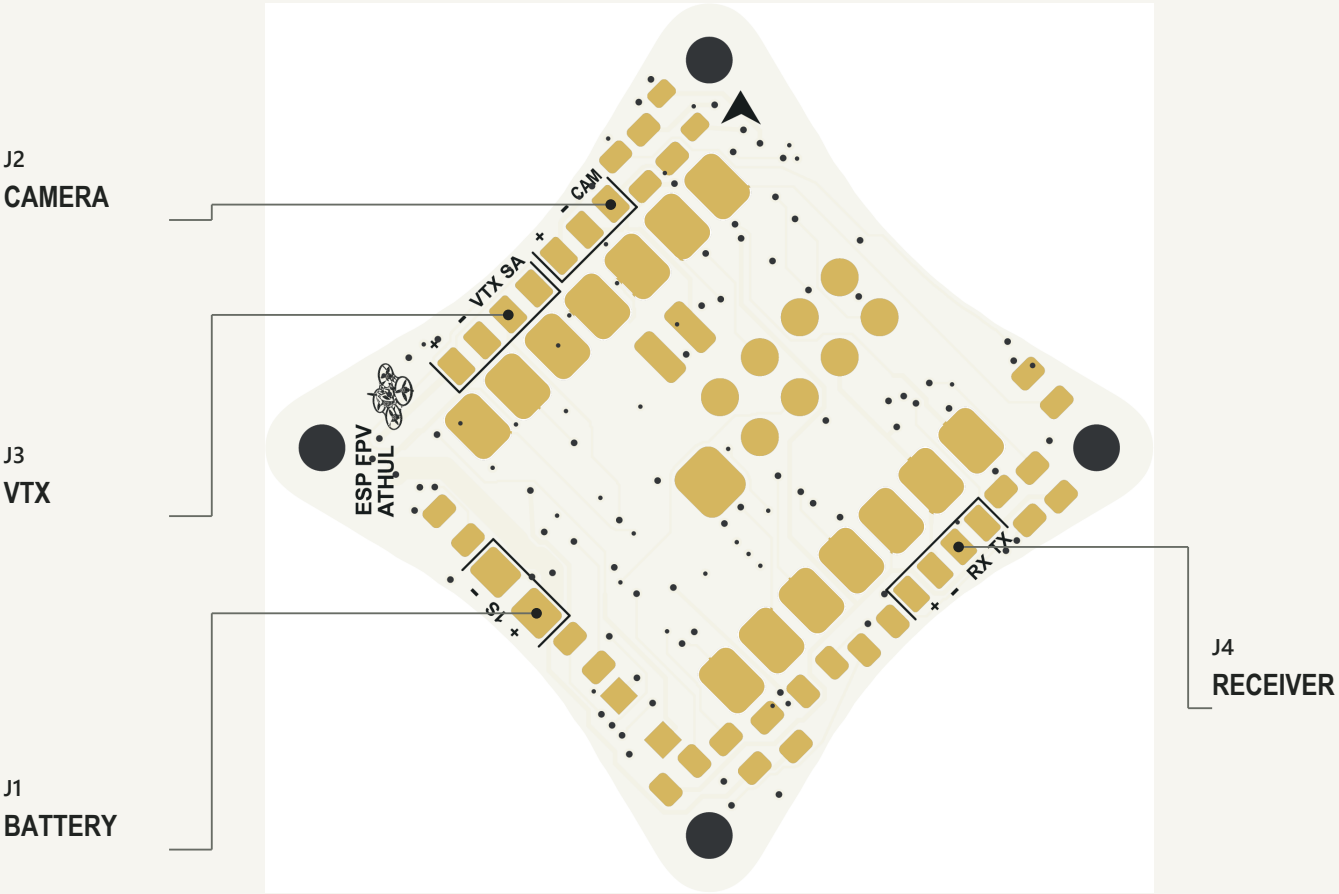
1S battery input with a boost stage for the assigned +5V peripheral rail.

1S	4	40 mm	USB
BATTERY INPUT	BRUSHED OUTPUTS	BOARD WIDTH	PROGRAMMING

02 CONNECT YOUR EQUIPMENT

EXTERNAL CONNECTIONS

Front side facing you. Numbers in the connection list are PCB pad numbers, not GPIO numbers.



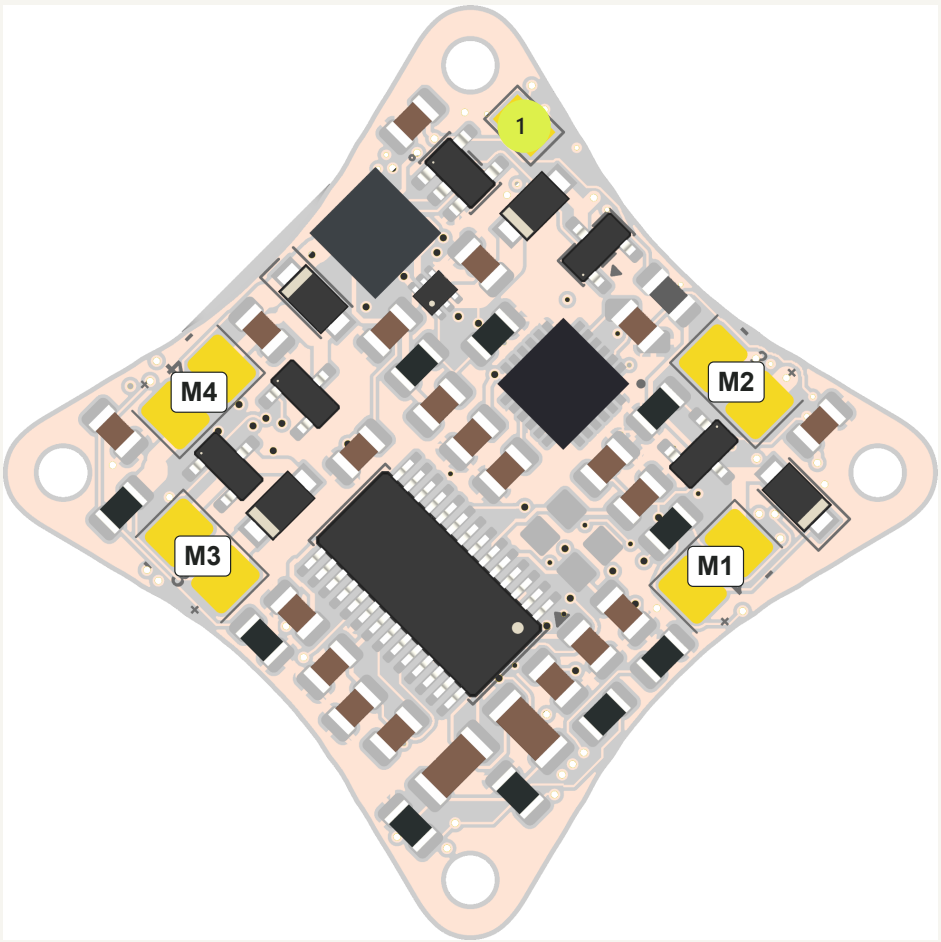
PORT	PAD ORDER	FUNCTION
J1 / BATTERY	1 VBAT+ / 2 GND	1S power input
J2 / CAMERA	1 +5V / 2 GND / 3 VIDEO	Video into OSD
J3 / VTX	1 +5V / 2 GND / 3 VIDEO / 4 SA	Video + SmartAudio
J4 / RECEIVER	1 +5V / 2 GND / 3 FC RX / 4 FC TX	Receiver serial link

> Receiver TX connects to FC RX; receiver RX connects to FC TX when needed. J2/J3/J4 power pads use +5V: check that each peripheral accepts 5V before connection. SA is a signal connection.

03 POWER & OUTPUTS

MOTOR OUTPUTS AND POWER

Four two-wire brushed-motor outputs. Use the PCB numbers when setting the mixer in firmware.



EACH MOTOR PORT

01 VBAT+

Positive motor supply

02 SWITCHED -

Controlled motor return.
This is not a ground pad.

BACK VIEW

Output labels identify PCB ports. They do not establish rotation or a verified flight mixer.

01 / JP1 POWER SELECTION

BATTERY OPERATION

JP1 closed. Use J1 for the battery input and check polarity before applying power.

USB + BATTERY

JP1 open when USB and battery are used together. Keep the battery unplugged during initial USB programming.

OUTPUT CHECK

Remove propellers. Test one output at a time, confirm the assigned GPIO and establish mixer order before flight.

04 FIRST USE

SETUP AND CONFIGURATION

Start with the assembled board electrically inspected. Keep propellers removed throughout setup.

STARTUP SEQUENCE

01 CHECK WIRING

Verify battery polarity and connections. Leave motors and external modules disconnected for first power.

02 VERIFY POWER

Use a current-limited supply for first power. Check 3.3V and +5V before connecting peripherals.

03 PROGRAM VIA USB

Keep the battery unplugged. Use firmware configured for this L4 board; no ready-to-fly firmware is supplied.

04 CHECK FUNCTIONS

Verify IMU axes, receiver input and OSD. Confirm PWM outputs start low; identify motors individually without propellers.

L4 / GPIO REFERENCE

IMU SDA	5
IMU SCL	43
Motor 1 PWM	1
Motor 2 PWM	2
Motor 3 PWM	39
Motor 4 PWM	6
FC RX / TX	40 / 41
OSD CS / SCK	3 / 7
SPI MISO / MOSI	8 / 9
SmartAudio	42
LED / battery ADC	44 / 4

GPIO NUMBERS, NOT FOOTPRINT PADS

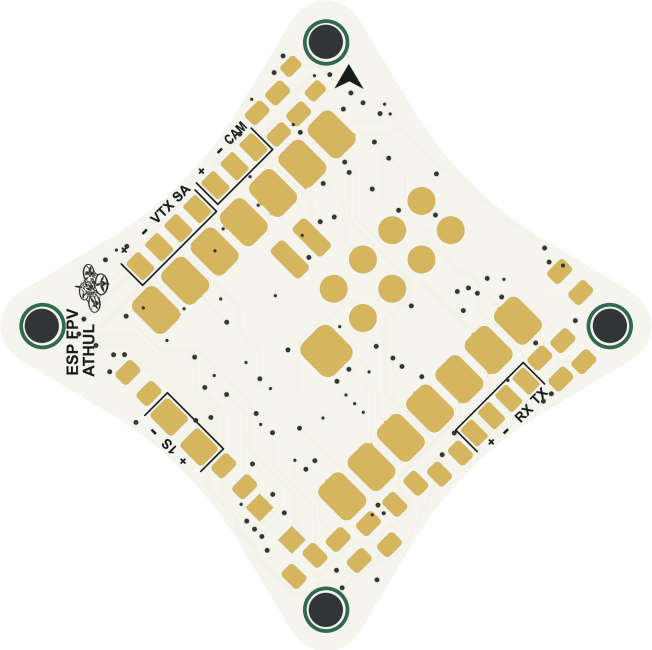
Firmware requirements. Set I2C SDA=5 and SCL=43 explicitly. Default XIAO SCL=GPIO6 is PWM4 here. Disable/remap UART0 TX on GPIO43 and initialize motor outputs low before motor power. Confirm the firmware uses the MPU-6050.

05 FIT & QUICK CHECKS

INSTALLATION AND SUPPORT

Check the mounting pattern and clearances before securing the controller.

DIMENSIONS / MM



40.00 overall

Outline	40 x 40 mm
Square hole pattern	24.65 x 24.65 mm
Opposite hole centers	34.857 mm
Mounting holes	4 x 2.10 mm

The hole square is rotated 45 degrees in this view. This is not a 25.5 mm pattern. Use insulating M2 hardware or isolating washers and leave USB access clear.

TROUBLESHOOTING

POWER / HEAT

Disconnect immediately if heating is unexpected. Check polarity, shorts and JP1 before resuming with limited current.

IMU NOT DETECTED

Check 3.3V, ground and I2C configuration: SDA5 / SCL43. Confirm the MPU-6050 driver.

NO VIDEO / OVERLAY

Check camera to J2, VTX to J3, common ground and peripheral voltage. Confirm OSD initialization.

NO RECEIVER INPUT

Cross TX/RX correctly. Check receiver protocol, +5V and FC RX40 / TX41.

WRONG MOTOR OUTPUT

Remove propellers. Check mixer order, PWM GPIO and switched-negative wiring. Test outputs individually.

L4 prototype hardware: assembled power, motor loading and flight behavior require validation. This manual covers connections and operation. Soldering details are in the separate Component Map.